

Systems Analysis and Design

Chapter 8: Requirement Definition in RUP



Learning Objectives

8.1 Definition, Deliverables, Types of Requirement

8.2 Problems & Risks

8.3 Process of Requirement Definition in RUP

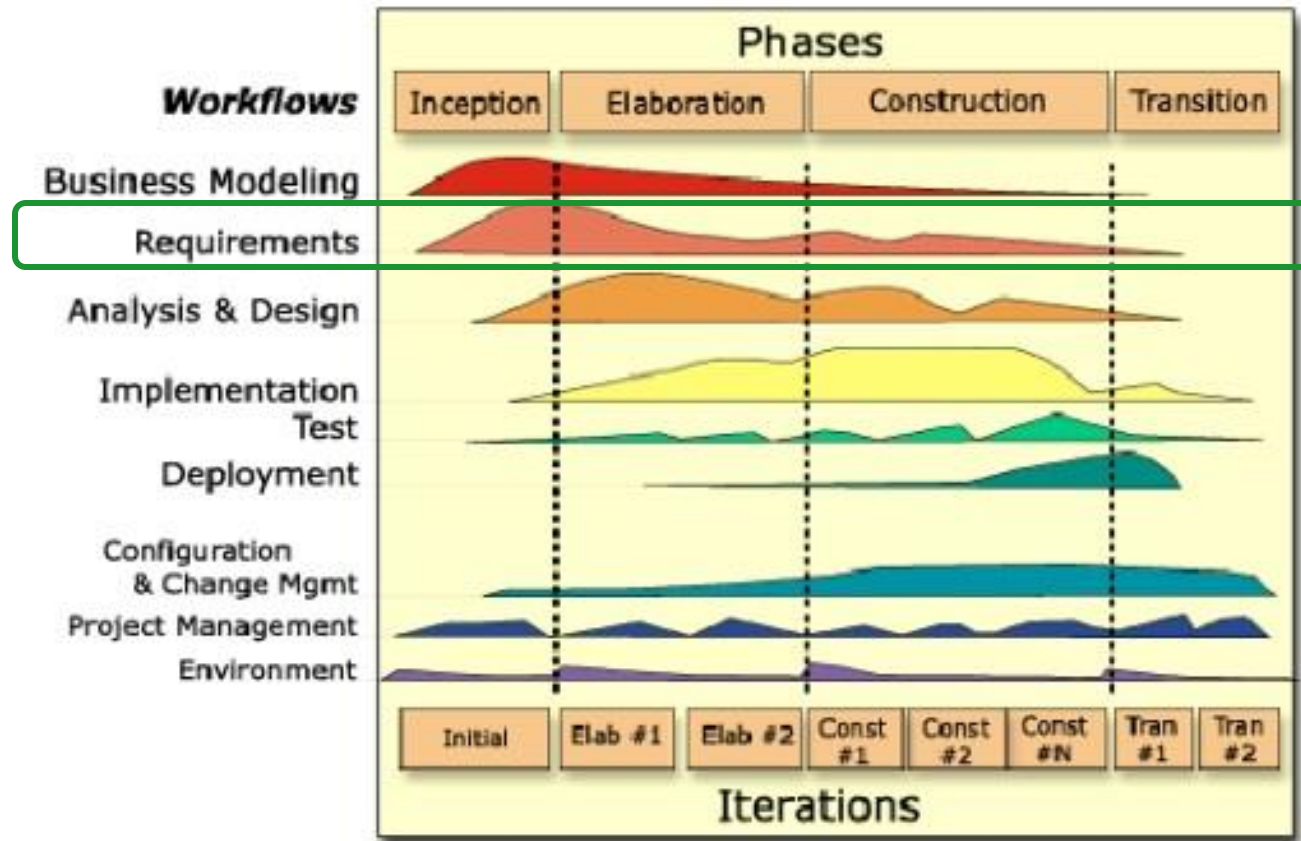
8.4 Requirement Specification

8.5 Requirement Analysis Techniques

8.6 Requirement Analyst, Business Analyst

Definition

- Requirement Definition (RD): *is a process of collecting the user needs to solve a problem or issues and **achieve** an objective*



Why?

- Only 16.2% of projects will be completed on time & on budget
- About **40-56%** of project conflicts can be traced to requirement errors
- Finding and fixing requirement errors consumes **70-85%** of project rework costs
- The average project exceeds its planned time schedule by **120%**
- About **52.7%** of projects will cost **189%** of their original estimate
- About **30%** of projects are cancelled before completion

Organizational Components to Understand (Chapter 3)

- Business objectives that drive what and how work is done
- Information people need to do their jobs
- The data (definition, volume, size) handled in support of jobs
- Data transformation and storage (when, how, by whom)
- Data handling dependencies and sequences
- Data handling and processing rules
- Policies and guidelines that describe the nature of the business and market and the environment it operates in
- Key events that affect data values and when they occur

Deliverables for Requirements Determination

(Chapter 3)

Deliverables for Requirements Determination	
1.	Information collected from conversations with or observations of users: interview transcripts, notes from observation, meeting minutes
2.	Existing written information business mission and strategy statements, sample business forms and reports and computer displays, procedure manuals, job descriptions, training manuals, flowcharts and documentation of existing systems, consultant reports
3.	Computer-based information: results from JAD sessions, reports of existing systems, and displays and reports from system prototypes

Types of requirement

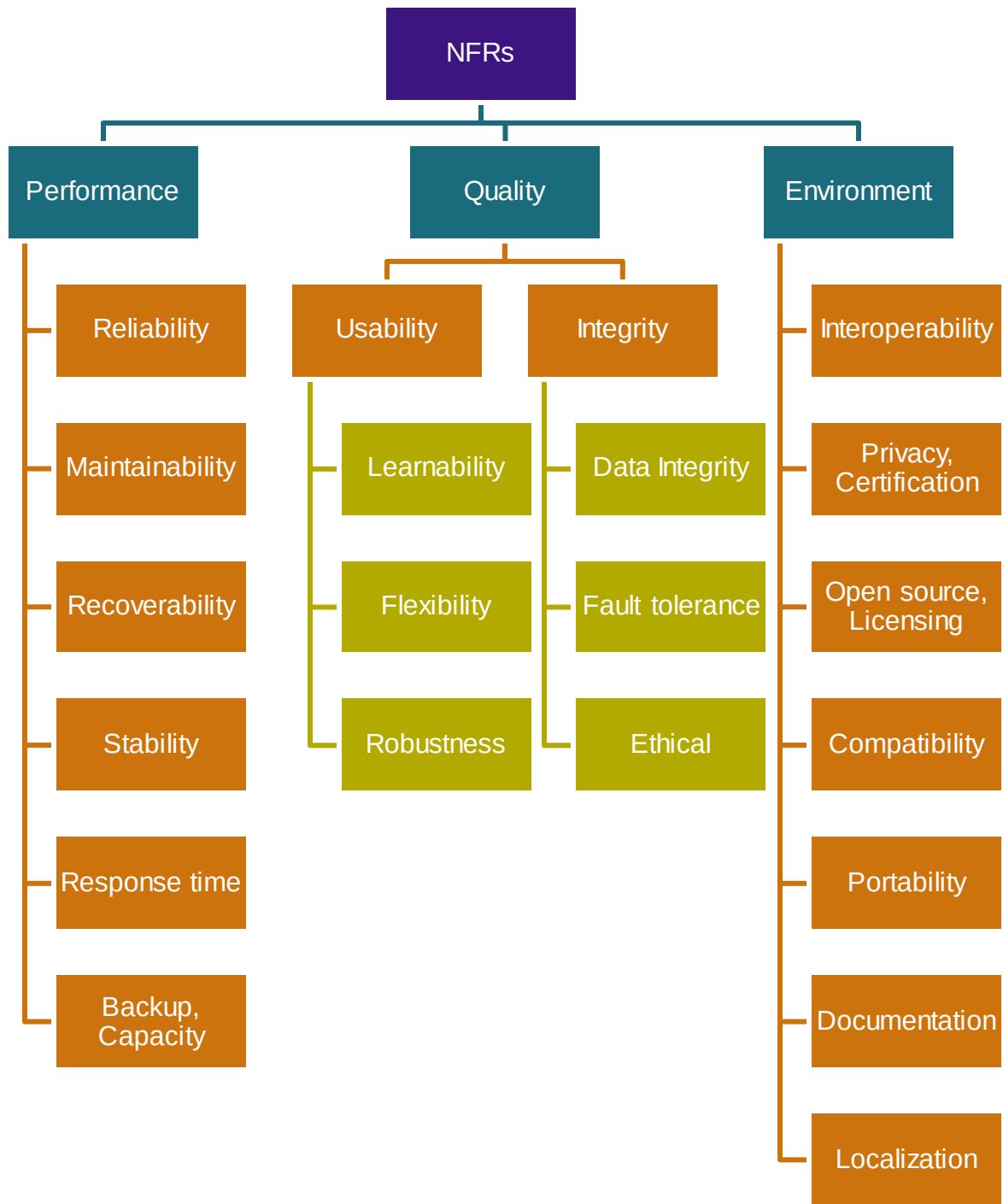
- Conscious Requirements
- Unconscious Requirements
- Undreamed Requirements
- Non Functional Requirements

Are those requirements that stakeholders are particularly aware of

Are those requirements that the stakeholder knows the requirement
exists but thinks that it's not worth mentioning

Talks about hardware and software requirement of an application. It also talks about application performance, number of normal and concurrent users, page response time

Non-functional requirement types



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Problems & Risks (1 of 2)

- Users have trouble in **explaining** what they want.
- Developers have trouble in **translating** user requirements into working programs and databases
- Both the business worlds and technology worlds are constantly **changing**

Problems & Risks (2 of 2)

- Scope and Vision not clearly defined
- All requirements are critical, no priority is defined
- Signed-off requirements keep changing
- New requirements get added in the middle of the project
- Users/ Customers are busy and not available to specify requirements
- Functionality built, rarely or never used

Requirement Volatility

- **Internal Factors**

- Not capturing requirements from all relevant stake holders
- Not using right requirement capturing technique depending on the context
- Not capturing all relevant details
- Not having measures such as check-list to ensure completeness of requirements captured
- Not having measures to ensure clarity

- **External Factors**

- Market driven
- Need to be handled and managed

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How the customer explained it



How the Project Leader understood it



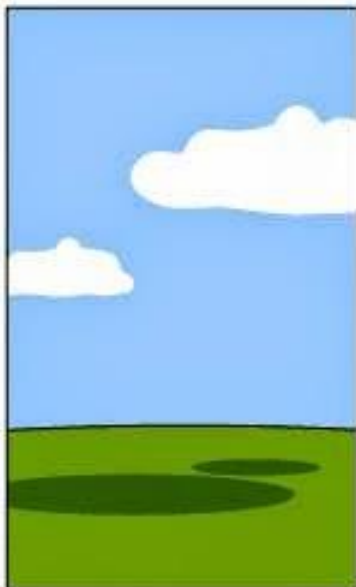
How the Analyst designed it



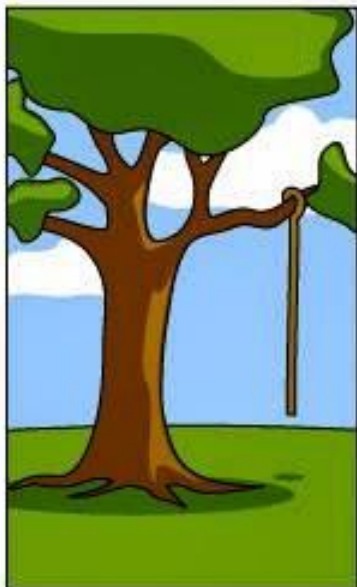
How the Programmer wrote it



How the Business Consultant described it



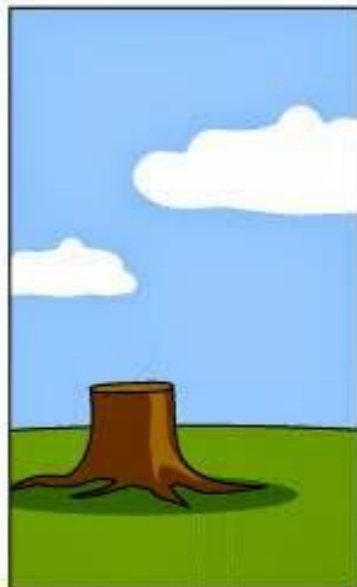
How the project was documented



What operations installed



How the customer was billed

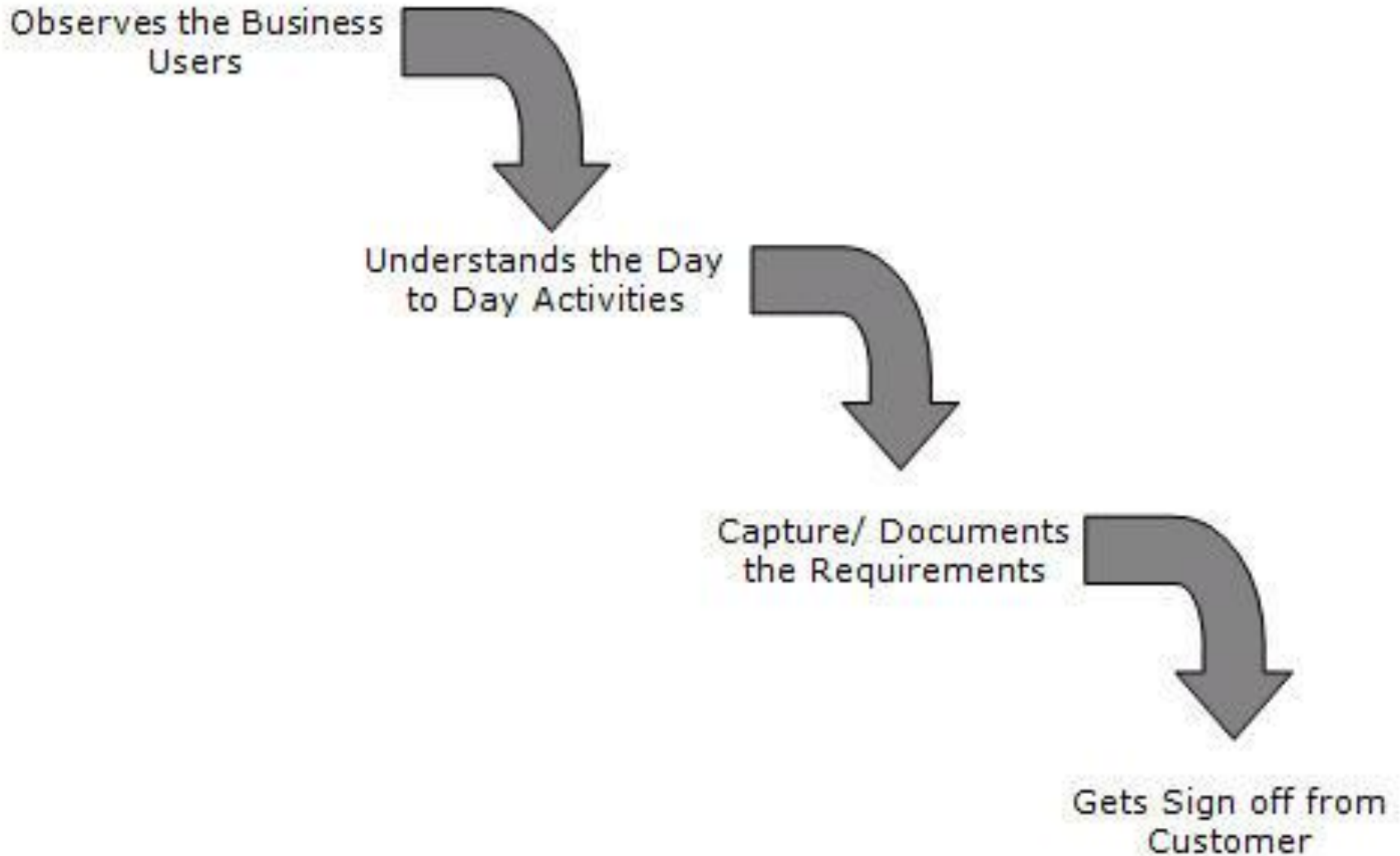


How it was supported

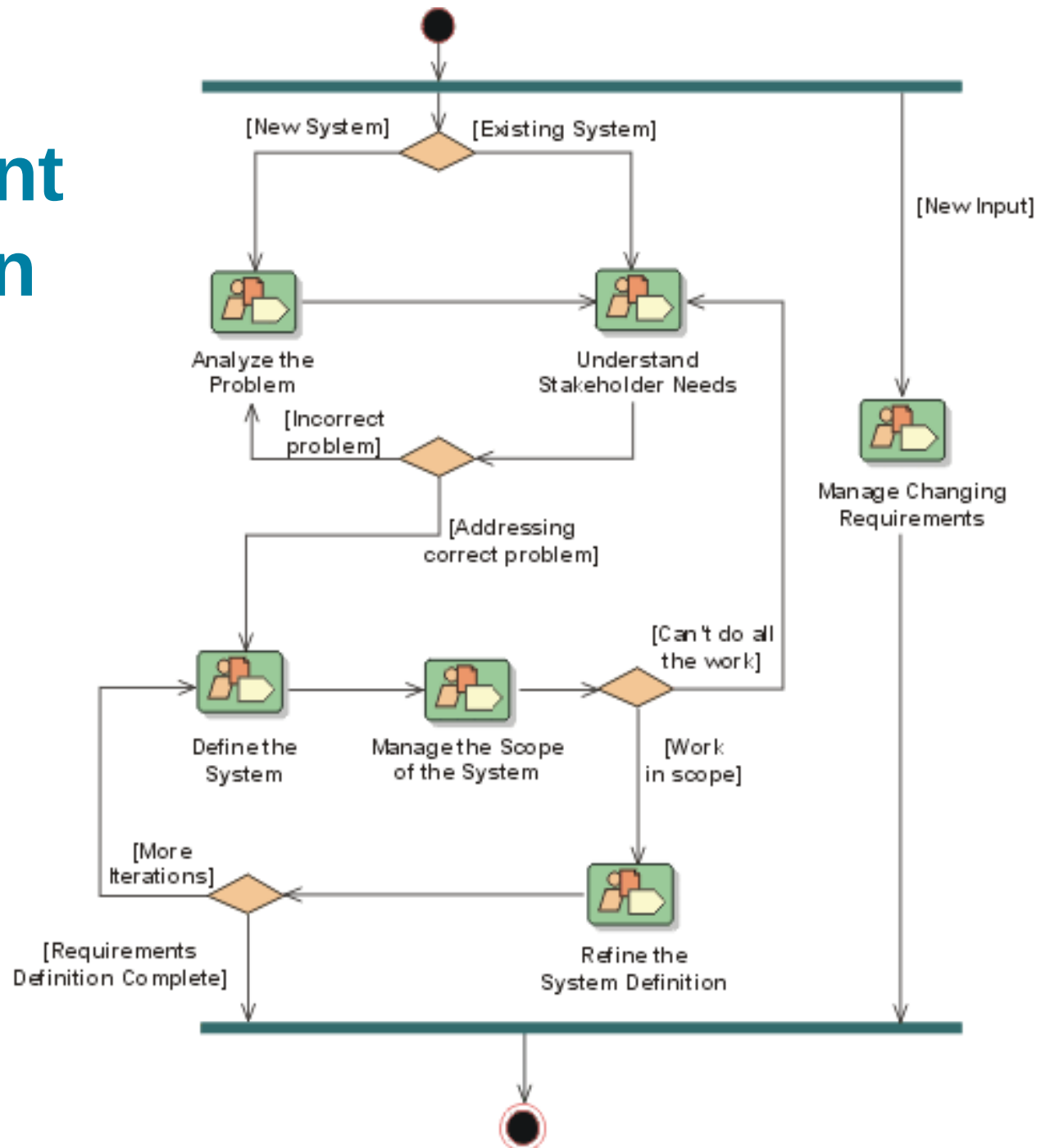


What the customer really needed

Process of Requirement Definition



Process of Requirement Definition in RUP



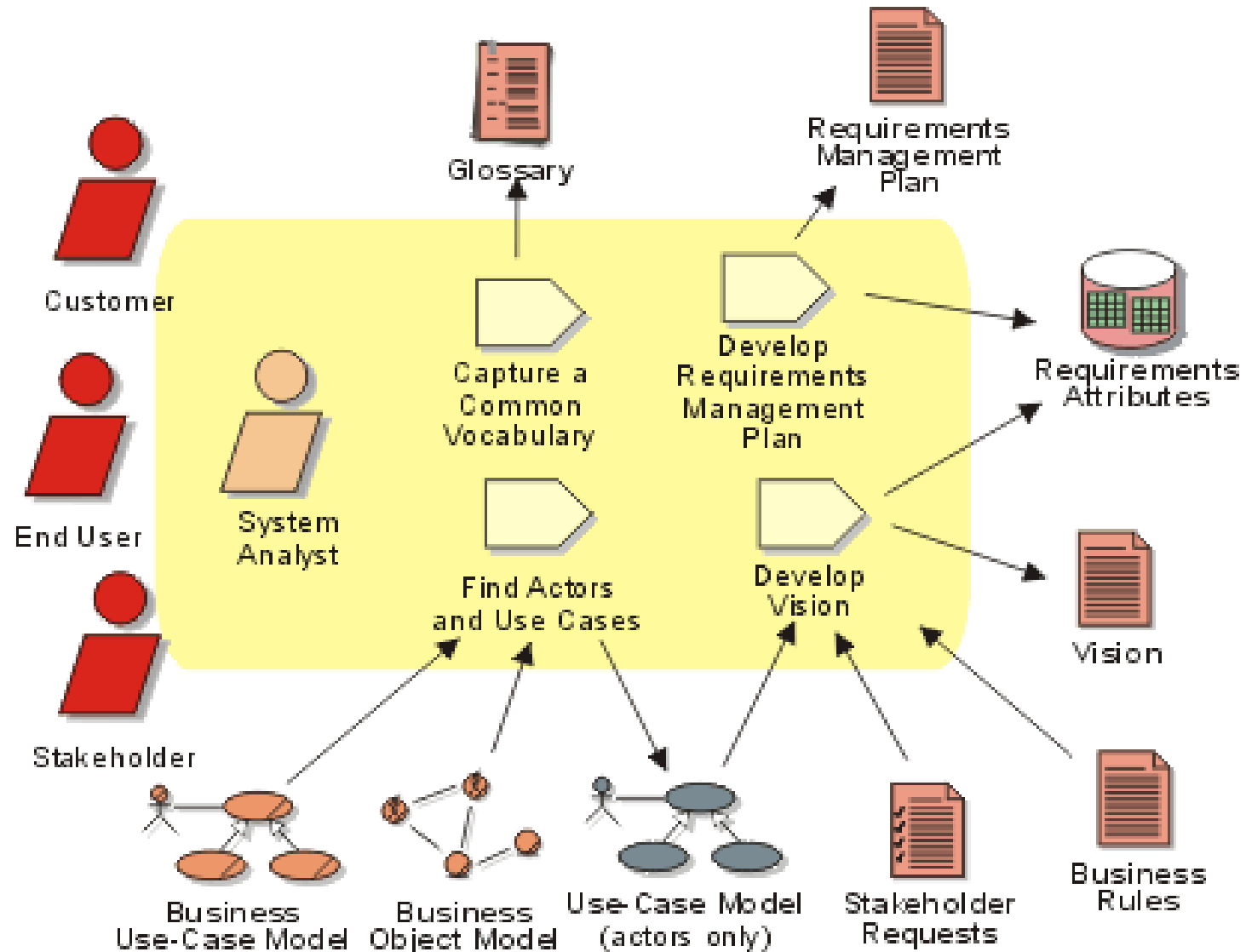
Managing the Scope of Project

- Carefully prioritize the requirements, based on input from all stakeholders, and manage its scope
- Carefully choosing requirements to for each increment that mitigates known risks in the project
- To do so, you need to negotiate the scope (of each iteration) with the stakeholders of the project. This typically requires good skills in managing expectations of the output
- Need to have control of the sources of requirements, of how the work products of the project look

Refining the System Definition

- The detailed definition of the system needs to be presented in such a way that your stakeholders can understand, agree to, and sign off on it
- Needs to cover not only functionality, but also NFRs
- Needs to produce different kinds of descriptions for different audiences
- Often in combination with simple visual prototypes
- Another component of the detailed definition of the system is to state how the system should be tested

Roles & Deliverables



The requirements document

- The requirements document is the official statement of what is required of the system developers
- Should include both a definition and a specification of requirements
- It is NOT a design document. As far as possible, it should set of WHAT the system should do rather than HOW it should do it

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Requirement Specification

- Glossary
- Requirements Management Plan
- Software Requirements Specification
- Supplementary Specification
- Stakeholder Requests
- Use-Case Specification
- Vision

Requirement Specification: Glossary

- The Glossary defines important terms used by the project
- Provides a consistent set of definitions to help avoid misunderstandings
- Project members initially use the Glossary to understand the terms that are specific to the project. This document is also important to people performing these roles:
 - **Developers** when designing and implementing classes, database tables, user-interfaces, and so forth
 - **Analysts**, can clearly define business rules, make correct and consistent use of those terms
 - **Developers** and **technical writers** construct training material and documentation using recognized terminology

Requirement Specification: Requirements Management Plan

- Describes how the project will set up:
 - requirements documentations
 - requirement types
 - and their respective requirements attributes and traceability
- Specifying the information and control mechanisms to be collected and used for measuring, reporting, and controlling changes to the product requirements
- The Requirements Management Plan is developed during the Inception phase and is updated at each major milestone

Requirement Specification: Software Requirements Specification

- Focuses on the collection and organization of all requirements surrounding your project
- Determines the correct location and organization of the requirements
- When using use-case modeling, this artifact consists of a package containing use cases of the use-case model
- The SRS package controls the evolution of the system throughout the development phase of the project

Requirement Specification: Supplementary Specification

- The Supplementary Specifications capture the system requirements that are not readily captured in the use cases of the use-case model. Such requirements include:
 - Legal and regulatory requirements, and application standards
 - Quality attributes of the system to be built, including usability, reliability, performance, and supportability requirements
 - Other requirements such as operating systems and environments, compatibility requirements, and design constraints
- The following roles and role sets use the Supplementary Specifications:
 - Developers: when define class responsibilities, operations, attributes
 - Managers refer for input when planning iterations
 - Testers use it to verify system compliance.

Requirement Specification: Stakeholder Requests

- Contains any type of requests a stakeholder (customer, end user, marketing person, and so on) might have on the system
- May also contain references to any type of external sources to which the system must comply
- Although the system analyst is responsible for this artifact, many people will contribute to it: marketing people, end users, customers
- This information may be collected in a document or automated tool and appropriate requests should be tracked and reported on following an approved Change Request Management (CRM) process.

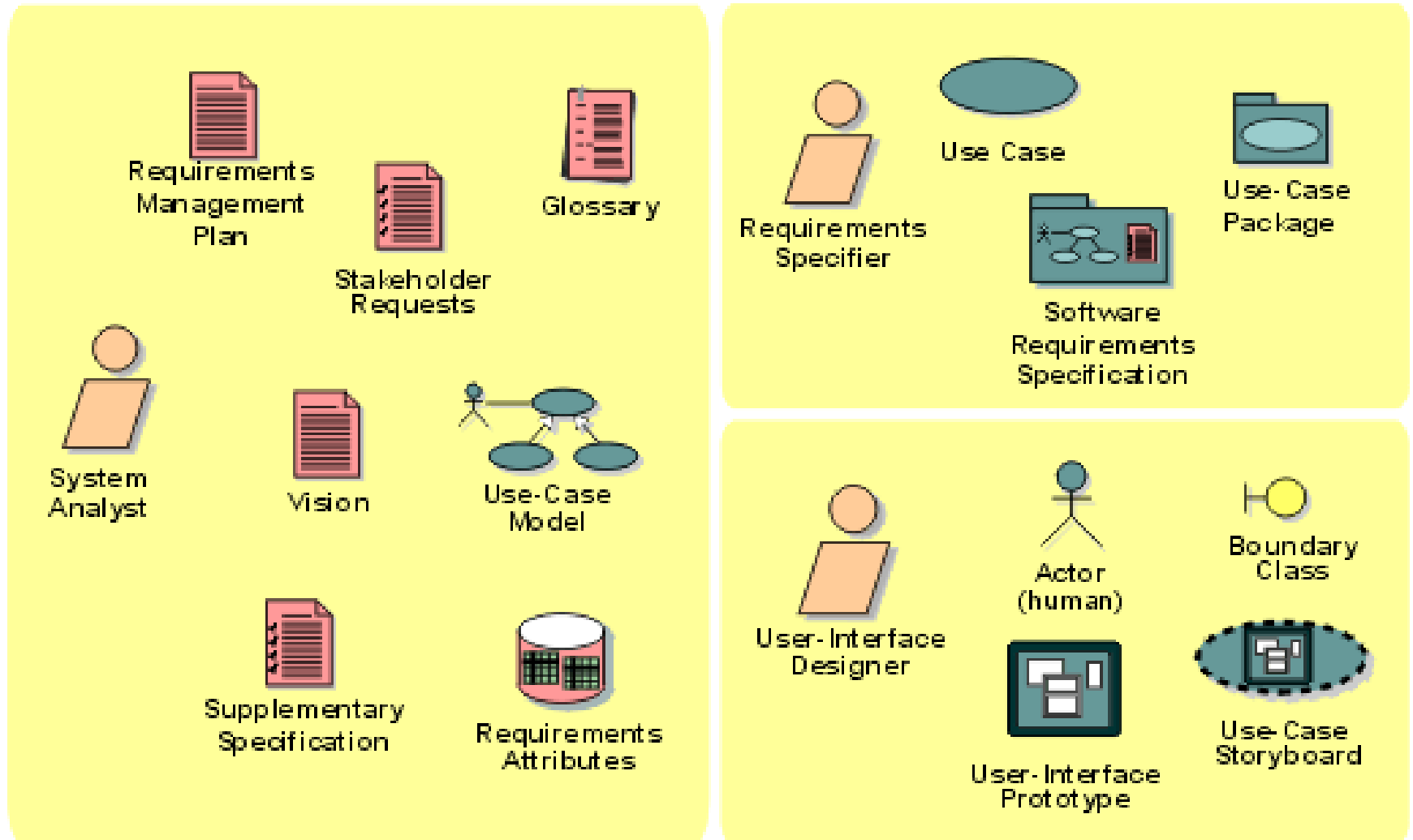
Requirement Specification: Use-Case Specification

- Describes what happens inside the system, if information is exchanged, be specific about what is passed back and forth
- Improves clarity by pasting graphical depictions of user interfaces, process flows or other figures into the use case
- Use case diagram, Activity diagram or even Sequence diagram often clarifies the behavior of a system better than pages upon pages of text
- Use the right presentation medium for your problem, but be wary of using terminology, notations or figures that your audience may not understand, remember that your purpose is to clarify, not obscure

Requirement Specification: Vision

- The Vision document provides a high-level—sometimes contractual—basis for the more detailed technical requirements
- The Vision captures very high-level requirements and design constraints to give the reader an understanding of the system to be developed
- It provides input to the project-approval process
- The Vision document will be read by managers, funding authorities roles in use-case modeling, and developers in general

Requirement Artifact Set



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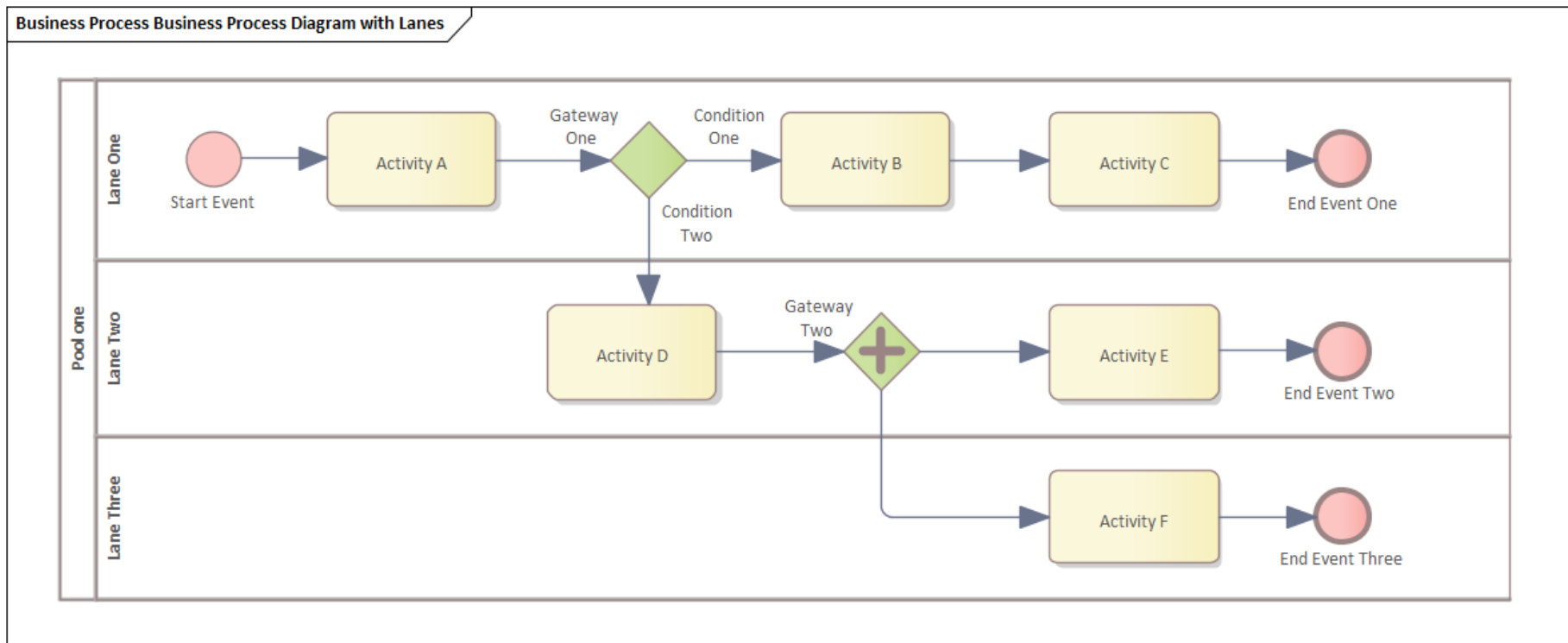
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Requirement Analysis Techniques

- Business process modeling notation (BPMN)



Requirements Validation

- Concerned with demonstrating that the requirements define the system that the customer really wants
- Requirements error costs are high so validation is very important
 - Fixing a requirements error after delivery may cost up to 100 times the cost of fixing an implementation error
- Requirements checking
 - Validity
 - Consistency
 - Completeness
 - Realism
 - Verifiability

Requirements Validation Techniques

- Reviews
 - Systematic manual analysis of the requirements
- Prototyping
 - Using an executable model of the system to check requirements.
- Test-case generation
 - Developing tests for requirements to check testability
- Automated consistency analysis
 - Checking the consistency of a structured requirements description

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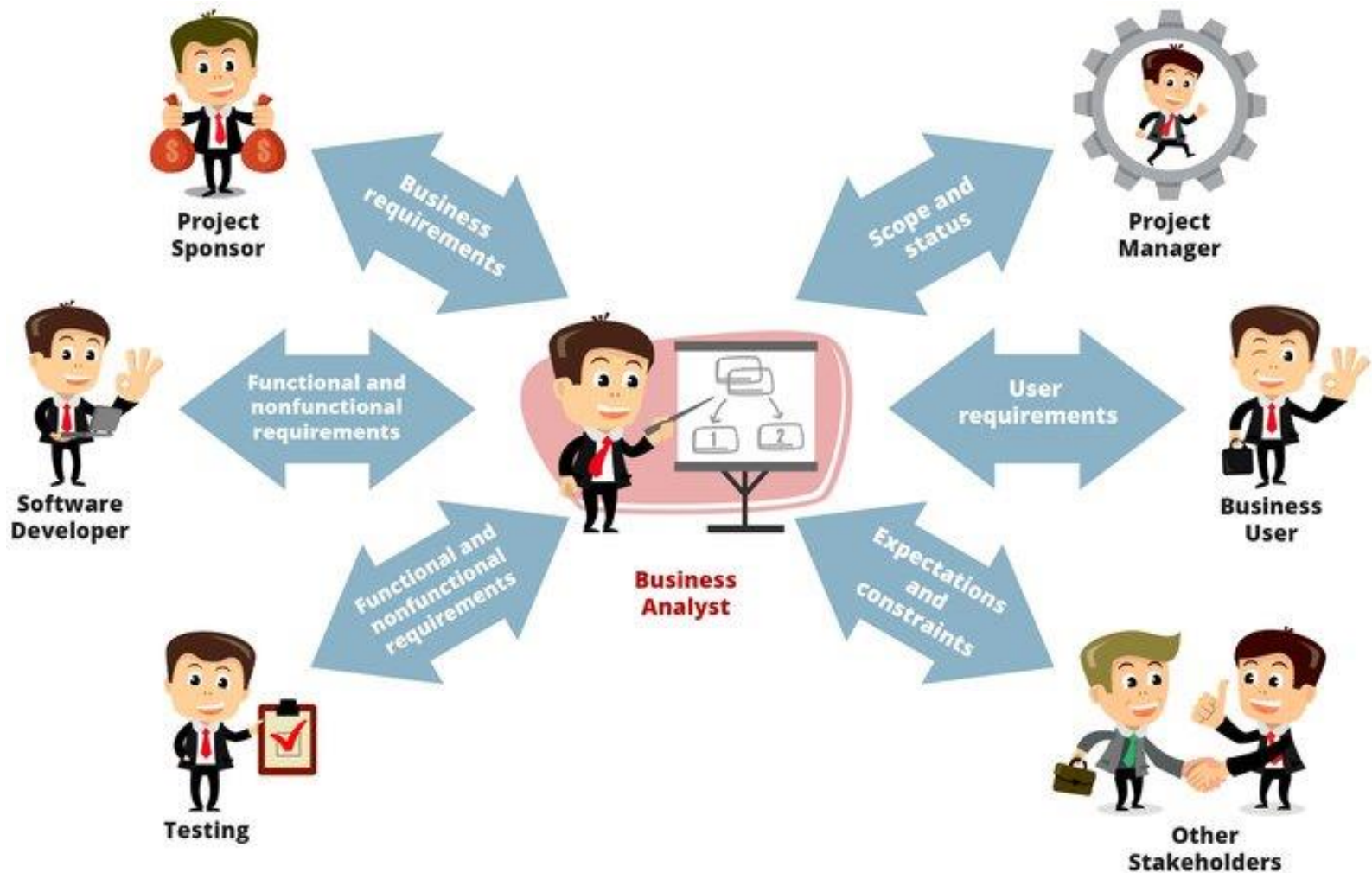
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Requirement Analyst

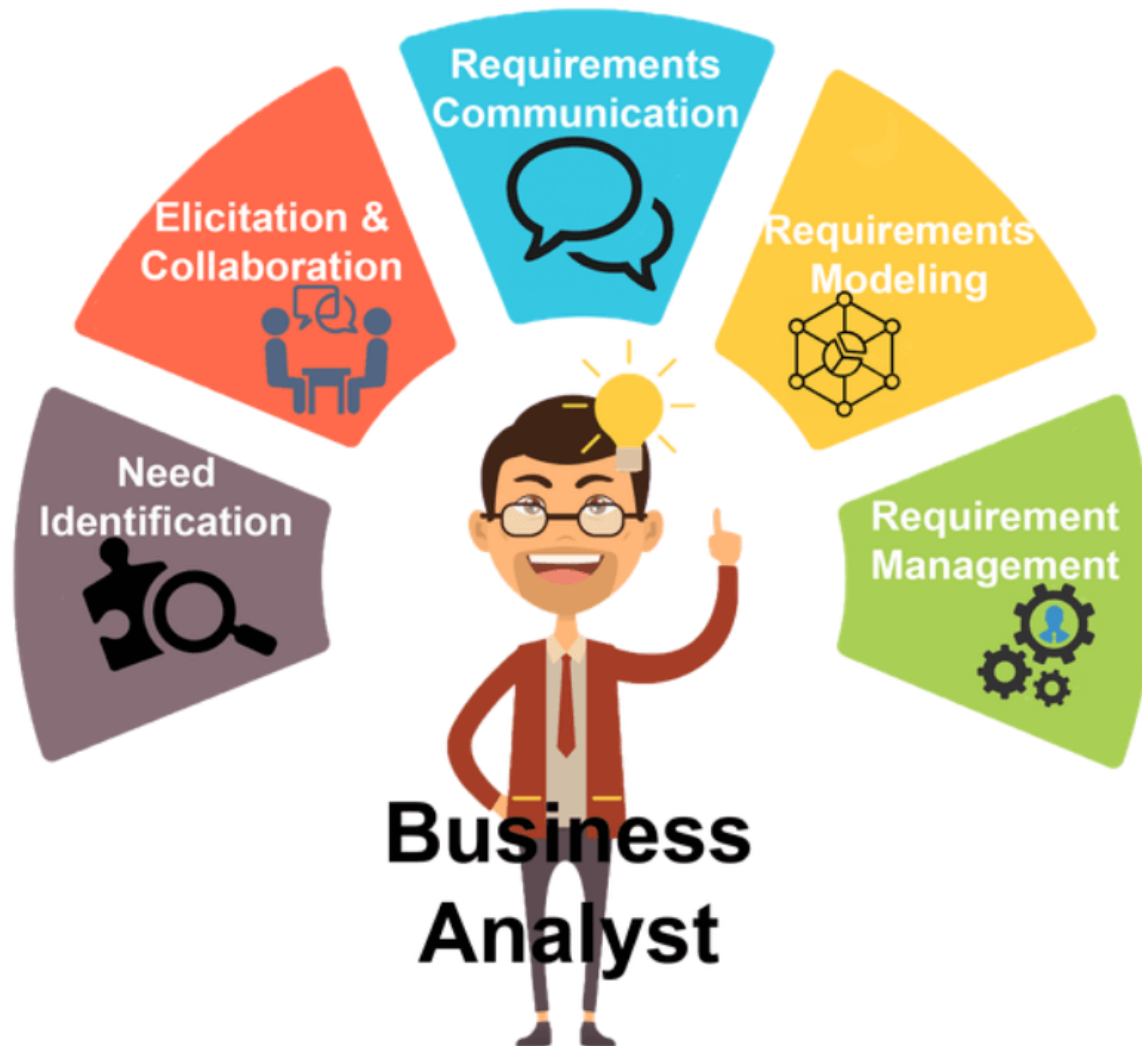
- Requirement Analyst (*or more well-known name: Business Analyst - BA*) describes business needs in the language understandable both for Business and for Development
- Building trust relationship with stakeholders
- Using various techniques and tools



Requirement Analyst: Communication



Requirement Analyst: Job Description



Summary

- Definition, Deliverables, Types of Requirement
- Problems & Risks
- Process of Requirement Definition in RUP
- Requirement Specification
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Question?